# Practice workbook · ASCII and Unicode character sets

How characters are encoded: ASCII bits and size, finding a code by counting on from a given code, benefits of ASCII, and why Unicode was needed.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-09

# 1. Question 2 · part a.iii · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a.iii only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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P 7 2 5 3 8 A 0 4 2 0

## 1. Question 2 · part a.iii · 2 marks · Working and answer

Working and answer

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# 2. Question 2 · part c.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.i only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

**(2)**

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

**(1)**

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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# 2. Question 2 · part c.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.i only.

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(c) Computers encode characters using ASCII and Unicode.

(i) State the number of characters that can be represented using standard ASCII.

**(1)**

(ii) The ASCII system uses 7 bits to represent a character. The ASCII code for the character 'A' using denary is 65; other alphabetical characters follow on from this in sequence.

Identify the ASCII code for 'H'.

**(1)**

- ☒ **A** 1000111
- ☒ **B** 1001010
- ☒ **C** 1001000
- ☒ **D** 1000101

(iii) Explain why Unicode was developed.

**(2)**

**(Total for Question 2 = 9 marks)**

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## 2. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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## 3. Question 2 · part c.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.ii only.

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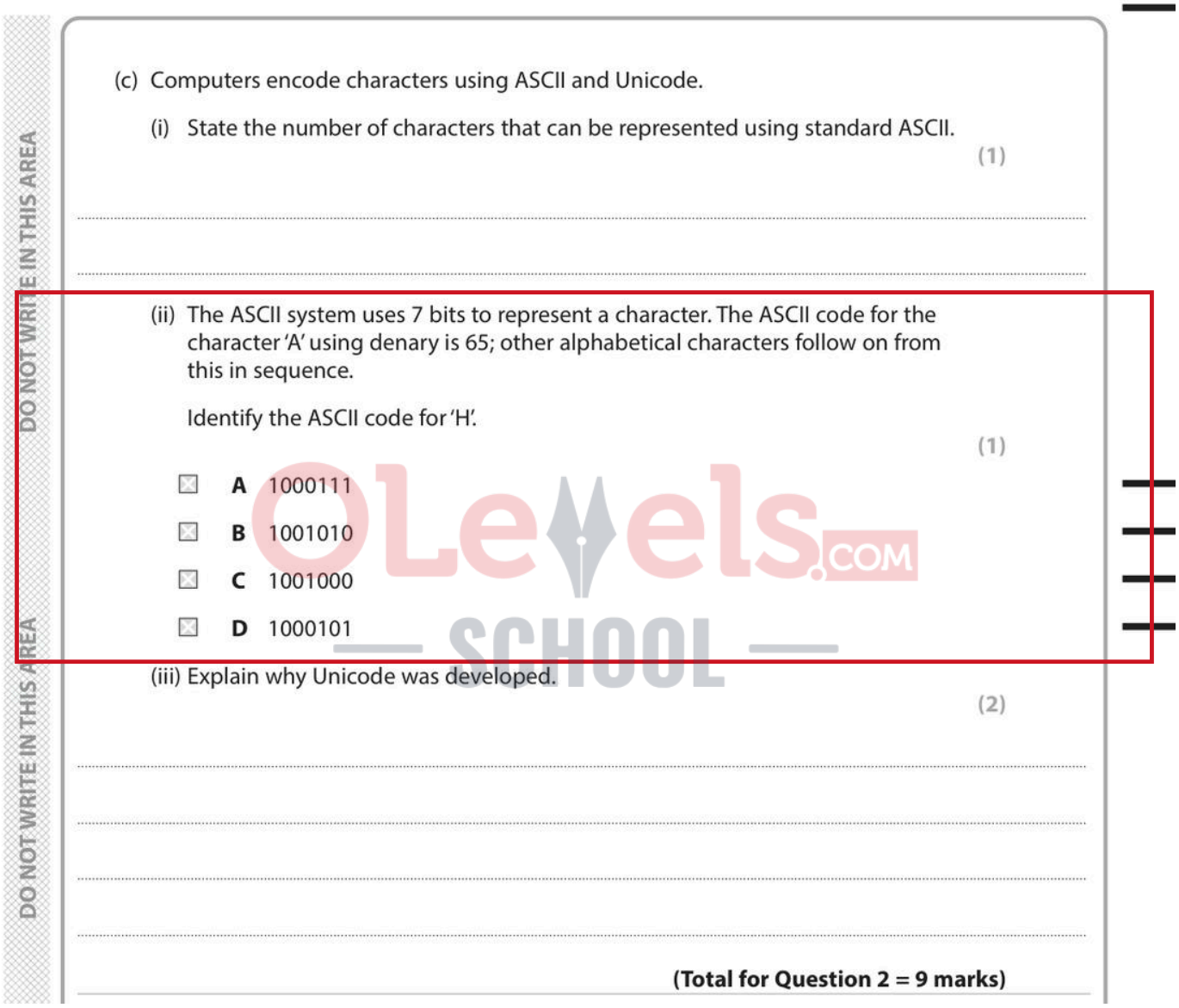
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## 3. Question 2 · part c.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.ii only.



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(iii) Explain why Unicode was developed.

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## 3. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part c.iii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c.iii only.

**2** Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

**(2)**

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Convert this signed integer to denary.

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# 4. Question 2 · part c.iii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c.iii only.

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.....

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**(Total for Question 2 = 9 marks)**

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## 4. Question 2 · part c.iii · 2 marks · Working and answer

Working and answer

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## 5. Question 1 · part f · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 5. Question 1 · part f · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f only.

(d) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| | | | | | | | |

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

**(1)**

.....

(f) Give the number of bits per character used by standard ASCII.

**(1)**

.....

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

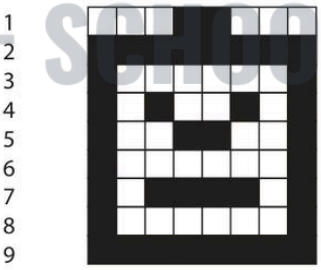



**Figure 1**

(i) State what is meant by the term **pixel**.

**(1)**

.....

(ii) Construct an expression to show the number of pixels in the image.

**(1)**

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## 5. Question 1 · part f · 1 mark · Working and answer

Working and answer

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# 6. Question 2 · part a.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

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## 6. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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## 7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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## 7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

**(2)**

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

**(1)**

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

**(2)**

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## 7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) The ASCII code for the character **1** is 011 0001

(i) Give the ASCII code for the character **5**

**(1)**

.....

(ii) Give **two** benefits of using ASCII to encode characters.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 3 = 13 marks)**

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Working and answer

## 8. Question 3 · part c.ii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part c.ii only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

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2024 JUN · Paper 01

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

**(2)**

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An error occurs.

Explain this error.

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2024 JUN · Paper 01

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Working and answer